

## **Lecture summary of Unit 3 Optics**

### **1. What is interference of light? What is constructive interference and what is destructive interference?**

Ans. This modification of average intensity due to superposition of two or more coherent light waves is known as the interference of light.

Interference is constructive if the amplitude of the resultant wave is greater than that of the individual interfering waves, and destructive if it is smaller than the individual waves.

### **2. What is meant by the term “coherent waves”? How the coherent waves are obtained in practice?**

Ans. Two sources are said to be coherent if they emit light waves of the same frequency, nearly the same amplitude and always have a constant phase difference between them.

Coherent sources are obtained by the following methods.

#### **1) Division of Wavefront**

The wavefront originating from a source of light is divided into two parts which serves the purpose of coherent sources. These two parts of the same wavefront travel unequal distances and reunite at some angle to produce interference bands.

e.g. Youngs’ double slit experiment.

#### **2) Division of Amplitude**

The amplitude of the beam is divided into two parts by partial reflection or refraction methods. The waves corresponding to the divided parts travel different paths and hence produce interference.

e.g. Interference due to thin films, wedge shaped film interference, Newton’s rings.

### **3. What is Optical path?**

Ans. Optical Path is the product of the geometrical path and the refractive index of the medium in which the light is travelling. Mathematically,

$$\text{Optical Path} = \text{Geometrical path} \times \text{refractive index of the medium.}$$

### **4. What are the conditions of sustained interference?**

Ans. The conditions for sustained interference are

1. The sources of light must be coherent.
2. The amplitudes of the two interfering waves should be equal or approximately equal in order to reduce general illumination.
3. Two sources should be very narrow as a broad source is equivalent to large number of

narrow sources lying side by side which causes loss of interference pattern resulting general illumination.

4. Two sources emitting set of interfering beams must be placed very close to each other.

**5. Explain the general conditions of interference maxima and minima (i.e. constructive and destructive interference).**

Ans. The condition for interference maxima is that the path difference between the two interfering waves must be an integer multiple of the wavelength ( $\lambda$ ) of the light. i.e.

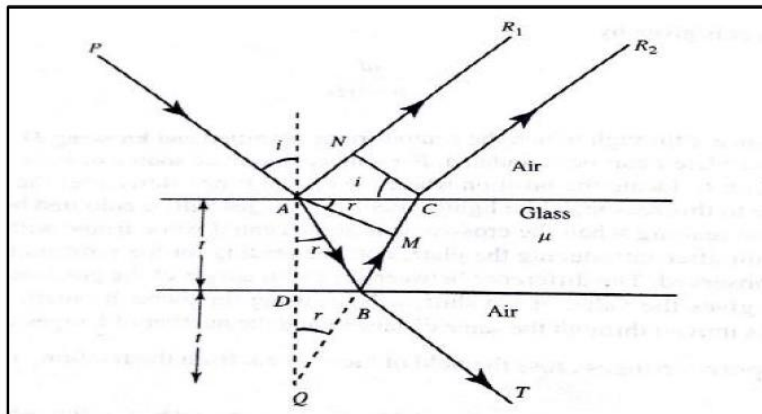
$$\text{path difference} = n\lambda \quad [n = 0, 1, 2, 3, \dots]$$

On the other hand, the condition for interference minima is that the path difference between the interfering waves is an odd number multiple of half wavelength ( $\lambda/2$ ). i.e.

$$\text{path difference} = (n + \frac{1}{2})\lambda \quad [n = 0, 1, 2, 3, \dots]$$

**6. Explain the interference in a thin parallel film and derive the conditions for its maxima and minima.**

Ans.



- Consider a thin transparent film of thickness  $t$  and refractive index  $\mu$ , as shown in the figure.
- A ray  $PA$  incident on the upper surface of the film is partly reflected along  $AR_1$  and partly refracted along  $AB$ . At the point  $B$ , a part of it is reflected along  $BC$  and finally after partial refraction it emerges out along  $CR_2$
- The difference in path between the two rays  $AR_1$  and  $CR_2$  can be calculated
- Draw  $CN$  normal to  $AR_1$  and  $AM$  normal to  $BC$ .
- The angle of incidence is  $i$  and angle of refraction is  $r$ .
- Also, produce  $CB$  to meet  $AQ$  produced at  $Q$ . Here,  $\angle AQC = r$ .

The optical path difference between the rays  $AR_1$  and  $CR_2$  is,

$$\Delta = \mu(AB + BC) - AN$$

Now from Snell's Law,  $\mu = \sin i / \sin r = (AN/AC) / (CM/AC) = AN/CM$

$$AN = \mu.CM$$

Therefore, the optical path difference,  $\Delta$  is

$$\begin{aligned}\Delta &= \mu(AB + BC) - \mu CM \\ &= \mu(AB + BC - CM) \\ &= \mu(QB + BC - CM) \quad [since, AB = QB] \\ &= \mu(QC - CM) \\ &= \mu QM\end{aligned}$$

In  $\Delta AQM$ ,  $\cos r = QM/AQ$

Therefore,  $QM = AQ \cos r = 2t \cos r$

$$QM = 2t \cos r$$

Hence, **the path difference,  $\Delta = 2\mu t \cos r$**

But, using **Stokes' Law in Optics**, we know that "a light ray reflected from a denser medium suffers a phase change of  $\pi$  or path difference of  $\lambda/2$ "

Therefore, the effective path difference is  $\Delta = 2\mu t \cos r \pm \lambda/2$

### **Condition for bright fringe**

*Path difference*  $= n\lambda$  or  $2\mu t \cos r \pm (\lambda/2) = n\lambda$ , where  $n = 0, 1, 2, \dots$

$$2\mu t \cos r = (2n \pm 1)(\lambda/2)$$

### **Condition for dark fringe**

*Path difference*  $= (2n+1)\lambda/2$  or  $2\mu t \cos r \pm (\lambda/2) = (2n+1)\lambda/2$ , where  $n = 0, 1, 2, \dots$

$$2\mu t \cos r = n\lambda$$

## **7. Describe the formation of color in soap film as seen under white light or sunlight.**

A. The formation of color in soap bubble is due to the interference of light from an extended source such as sky (sun). The films are usually observed by reflected light.

The eyes looking at the thin films receive light waves reflected from top and bottom surfaces of the thin soap film. The reflected rays are very close to each other and are in a position to interfere. We know that the condition for interference maxima is given by

$$2\mu t \cos r = (2n \pm 1)(\lambda/2) \quad n = 0, 1, 2, 3, \dots$$

It is seen that the above equation depends upon the thickness ( $t$ ) of the film, the wavelength ( $\lambda$ ) and the angle of refraction ( $r$ ), which is related to the angle of the incident light on the film.

White light consists of several (seven) wave lengths and for specific values of  $t$  and  $r$ , waves of only certain wavelength interfere to satisfy the above condition of constructive interference. Therefore, only those colors are present in the reflected light. Different colors will satisfy the condition at different angles. The other wave lengths interfere destructively and hence are absent in the reflected light. Hence, the film at particular point appears colored.

**8. If energy is always conserved, how come the interference maximum has amplitude more than the sum of individual interfering rays?**

Ans. The light energy which is absent in the dark fringes reappears as additional energy in the bright fringes. However, the total energy in the dark and the bright fringes remains same. Therefore, there is no violation in the conservation of energy.

**9. Write the expression for fringe width in a wedge shaped thin film.**

Ans. Fringe width  $\beta = m\lambda/2\mu\theta$  where,  $m$  = order of interference,  $\lambda$  = wavelength of the light,  $\mu$  = refractive index of the medium, and  $\theta$  = angle of the wedge.

For  $m = 1$ ,  $\beta = \lambda/2\mu\theta$

**10. State the relation between the phase difference and path difference.**

Ans. If the path difference between the two waves is equal to the wavelength,  $\lambda$ , the corresponding phase difference is  $2\pi$ .

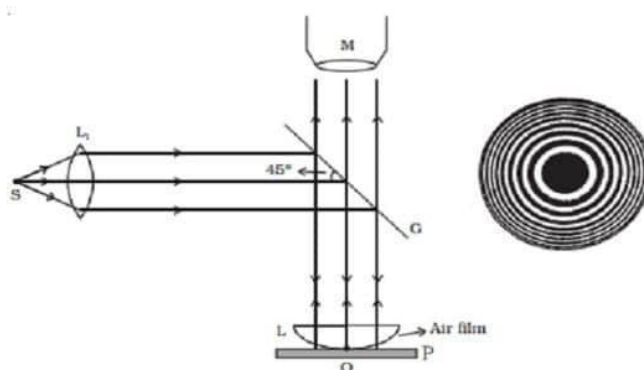
Hence, for path difference  $x$ , the phase difference,  $\delta = 2\pi x/\lambda$

Or

Phase difference,  $\delta = (2\pi/\lambda) * (\text{path difference})$

**11. What are Newton's rings?**

Ans. When a plane convex lens of long focal length is placed over an optically plane glass plate, a thin air-film of varying thickness is enclosed between them. The thickness of the film is zero at the point of contact, and gradually increases as we go outward. When the airfilm is illuminated with a monochromatic light normally, alternate bright and dark rings are formed with dark spot at the center. These rings are known as Newton's rings.



**12. In the Newton's ring experiment prove that in reflected light the radius of the dark rings is proportional to the square root of its order number.**

Ans.

Let us consider the vertical section SOP of the plano convex lens through its centre of curvature C, as shown in Fig . Let R be the radius of curvature of the plano convex lens and O be the point of

contact of the lens with the plane surface. Let t be the thickness of the air film at S and P. Draw ST and PQ perpendiculars to the plane surface of the glass plate. Then ST = AO = PQ = t

Let  $r_n$  be the radius of the nth dark ring which passes through the points S and P.

Then SA = AP =  $r_n$

If ON is the vertical diameter of the circle, then by the law of segments

$$SA \cdot AP = OA \cdot AN$$

$$r_n^2 = t(2R - t)$$

$$r_n^2 = 2Rt \text{ (neglecting } t^2 \text{ comparing with } 2R)$$

$$2t = \frac{r_n^2}{R}$$

According to the condition for darkness

$$2t = n\lambda$$

$$\frac{r_n^2}{R} = n\lambda$$

$$r_n^2 = nR\lambda \text{ or } r_n = \sqrt{nR\lambda}$$

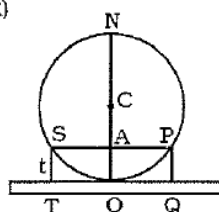


Fig . Radius of Newton's rings

Since R and  $\lambda$  are constants, we find that the radius of the dark ring is directly proportional to square root of its order. i.e.  $r_1 \propto 1$ ,  $r_2 \propto$

2,  $r_3 \propto 3$ , and so on. It is clear that the rings get closer as n increases.

**13. What is diffraction of light? Discuss different types of diffraction phenomenon. (or, Differentiate between Fresnel and Fraunhofer diffraction patterns).**

Ans. The bending of light when it encounters an obstacle or an opening that is of the order of the wavelength of the light is called diffraction.

Diffraction phenomenon can be classified into following two general classes:

**FRESNEL'S DIFFRACTION**

- The source and the screen are placed at finite distances from the aperture of the obstacle having sharp edges.
- The incident wavefronts are either spherical or cylindrical.

**FRAUNHOFER'S DIFFRACTION**

- The source and the screen are placed at infinity or optically at infinity.
- The incident wavefront is plane.

**14. State the conditions for (i) diffraction minima for single slit and (ii) diffraction maxima for multiple slits diffraction grating.**

Ans. The condition for diffraction minima for a single slit is given by

$$b \cdot \sin\theta = n\lambda \quad \text{where } b = \text{width of the slit}$$

The Condition for diffraction maxima for grating is given by

$$d \cdot \sin\theta = n\lambda$$

where,  $d$  = grating constant  $= (a+b)$

here,  $a$  and  $b$  are the width of transparent and ruled portion of the grating, respectively

$\theta$  = angle of diffraction

$n$  = order of diffraction

$\lambda$  = wavelength of the light used

**15. Write short notes on (i) dispersive power and the (ii) resolving power of a grating.**

Ans. The dispersive power of a grating is defined as the rate at which the angle of diffraction varies with wavelength. The factor  $(d\theta/d\lambda)$  is called the dispersive power for that order.

The diffraction of the  $n^{\text{th}}$  order principal maximum for a wavelength  $\lambda$  is given by

$$d \cdot \sin\theta = n\lambda$$

where,  $d$  is the grating element,  $n$  is the order of diffraction and  $\theta$  is the angle of diffraction

Differentiating the above equation w.r.t.  $\lambda$ , we get,

$$d \cos\theta \, d\theta = n \, d\lambda$$

or

$$d\theta/d\lambda = n / d \cos\theta$$

The Resolving power of grating is defined as the ratio of the wavelength of any spectral line to the difference in the wavelength between this line and a neighbouring line such that the two lines appear to be just resolved.

Thus, *the resolving power of grating = order of spectrum  $\times$  total number of lines on grating*

Therefore, resolving power,  $\frac{\lambda}{\Delta\lambda} = mN$ , where  $m$  = order of diffraction and  $N$  = total number of lines on the grating.